

Undergraduate Committee Report: Faculty Meeting, March 17, 2015

1. **Faculty vote on LAS changes to Math and Chemistry courses: Effective Fall 2016**

- a. Summary of Proposed Changes (details on next page):
 - i. Math 180/181 → 5 to 4 credit hours (-2 credit hours)
Adding vectors, matrices to course curriculum
 - ii. CHEM 112 → CHEM 122/123; CHEM 114 → CHEM 124/125 (splitting lecture/lab)
(No change in course curriculum or credit hours for Gen Chem)
 - iii. CHEM 233 (Organic Lab) → 1 credit hour to 2 credit hours (+1 credit hour)
- b. CHE program impact:
 - i. "Electives Outside of Major Rubric" hours increased from 3 to 4
Issue- most courses available are 3 credit hours; to satisfy requirement, students would have to take 2 courses.
 - ii. "NonEng and Gen Ed degree requirements" hours decreased from 74 to 73.
- c. Recommendations:

Option 1:

 - i. Keep "Electives Outside of Major Rubric" at 3 credit hours
 - ii. Change CHE 397 from 3 to 4 credit hours; Change Total Hours Required in College of Engineering to 49 hours
 - iii. "NonEng and Gen Ed degree requirements" hours decreased from 74 to 73.

Option 2

 - i. Keep "Electives Outside of Major Rubric" at 3 credit hours
 - ii. Change ENGR100 from 0 to 1 credit hour; Change Total Hours Required in College of Engineering to 49 hours: : (Being discussed at COE level)
 - iii. "NonEng and Gen Ed degree requirements" hours decreased from 74 to 73.

2. **ABET rubrics**

- a. **REMINDER** to complete Fall 2014 rubric for sample of students in courses (30 students)

3. **Senior Exit Interviews/Oral Exam**

- a. DRAFT SCHEDULE for student interviews/oral exams in April:
 - i. April 6-10, April 13-17, April 20, 22, 24, April, 27, 29, 30
 - ii. **Please send me what dates/times work for you**
 - iii. Faculty will interview 5 students each; Alan(12 students)

4. **Undergraduate recruiting update**

2015-Mar: 220 applications, 150 admitted, 5 accepted
2014-Mar: 247 applications, 141 admitted, 4 accepted
2013-Mar: 211 applications, 111 admitted, 12 accepted
2012-Mar: 174 applications, 86 admitted, 13 accepted

Recruitment/Enrollment

- a. ITE students (Accepts): Thank you emails have been sent out
- b. 10 ChE Women in Engineering Scholarships: Congratulatory emails sent to students
 - i. 3 Unger awards, 1 Unger + Engineering Merit Awards, 1 Major + Engineering Merit Awards, 2 Dean's Scholarship Awards, 3 SWE Knowles Scholarships

5. Teaching/lab equipment:

- a. Process control system; ready for CHE 341
- b. Distillation column setup ongoing
- c. Classroom modules-
 - i. Tube/shell Heat exchanger shown to CHE201 class (no demo)
 - ii. Heat exchanger demo planned for CHE 312j

6. CHE Career Initiatives Task Force

- a. UIC Chemical Engineering LinkedIn account- existing since 2012 (Han Wang owner; contacted Han and he wants to help with updating LinkedIn acct and alumni/student outreach efforts)
- b. ChE alumni list with company contacts initiated by Rose Coppola-Conroy

Respectfully submitted,
Alan Zdunek, DUGS

LAS Justification for Changes

CHEM 112/ CHEM 114/CHEM 233

Separating the lecture and laboratory components of general chemistry courses into concurrent lecture and laboratory courses provides a better structure to support student progress and course accessibility. Currently, CHEM 112 (and 114) are comprised of a lecture and laboratory component; students complete a single registration and receive one grade. Because the courses award five hours of credit, poor performance in one component of a course (especially lecture) has a significant impact on the student's GPA; and the student who wishes to improve his/her grade must retake the entire course. With the revision, students who do well in, for example, the laboratory course, but not the lecture course, will not need to repeat both courses. This revision will also benefit the student's GPA because it will lessen the impact of a poor grade, while and permitting the inclusion of the student's better grade.

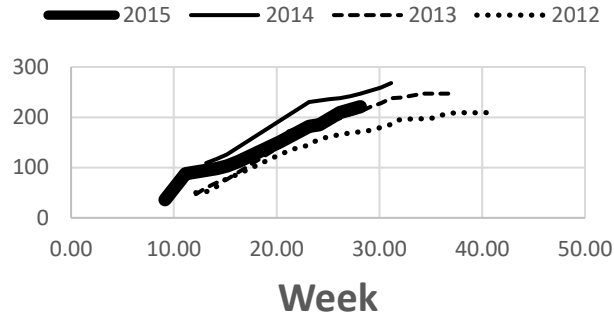
CHEM 233, Organic Chemistry, is a laboratory course that meets for 4 hours per week but awards only 1 hour of credit. There are no other chemistry courses with this many lab contact hours for only one hour of credit. The increase in credit hours to 2 will bring the course into consistency with the campus' credit hour/contact hour formula.

MATH 180/ MATH 181/ MATH 210

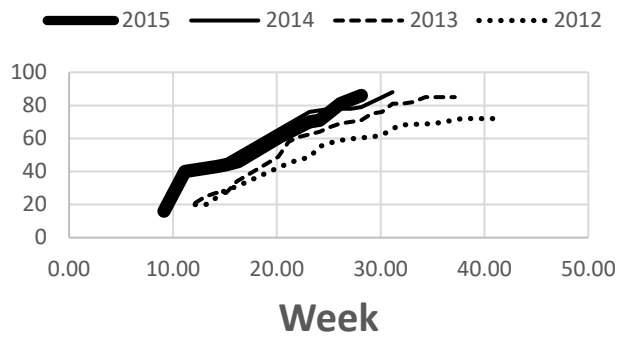
At most universities, the calculus I—calculus II sequence is, at most, 8 credit hours. Although the number of credit hours for Math 180 (and 181) will be reduced, the number of contact hours is unchanged and students will continue to have three hours of lecture and 2 hours of discussion each week. Discussion sections in Math 180 and Math 181 are problem sessions where students engage in active group learning projects under the direction of teaching assistants. As no additional material is covered in these sessions it is reasonable that the number of credit hours is lower than the number of contact hours. At the request of stakeholder departments, the rudiments of linear algebra are being included in the first-year calculus sequence. A section on vectors will be added to Math 180 (and a section on matrices to Math 181). The addition of this material should help students in courses like Math 220 (Differential Equations) and many courses in the physical sciences and engineering. To make room for this material, some of the most technical material on techniques of integration is being removed.

Math 210 previously had 1 hour per week of laboratory-discussion devoted to computer methods in multivariable calculus. Changes in technology make it much easier to incorporate this material seamlessly in the lectures. At the same time, the department feels that students were suffering by not having enough time devoted to problem solving. For this reason, the lab-discussion is being changed to a problem session where students engage in active group learning activities under the direction of a teaching assistant.

Applicants - All



Applicants - Women



Applicants - Minority

